Geometric Group Theory

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Lecture 1

1 Combinatorial group theory

Combinatorial group theory is a sibling field to Geometric group theory. Both study infinite discrete groups.

1.1 Free groups and presentations

Let $A = \{a_1, a_2, \dots\}$ be an *alphabet*. A group F is *free on* A if

1. There is a map of sets $A \to F$,
2. for any group G, and a map of sets $A \to G$, there exists a unique group homomorphism $F \to G$ such that

$$\begin{array}{ccc} & & F \\ & \nearrow & \downarrow \\ A & \longrightarrow & G \end{array}$$

commutes.

This is a *universal property*. As usual, this means that F is unique up to unique isomorphism. This shows that F is determined by A, so we may write $F = F(A)$.

However (as usual with definitions by universal property), we don't know if $F(A)$ exists. We'll show this two different ways.

1. Topologically: Let

$$X = \bigvee_{a \in A} S^1$$

By the Seifert-van Kempen theorem,

$$\pi_1(X) \cong F(A)$$

2. Combinatorially: Let

$$A^* = \{\text{words in } A \sqcup A^{-1}\}$$

where $A^{-1} = \{a_1^{-1}, \dots, a_n^{-1}\}$. For example,

$$1 = \varnothing, aa, aa^{-1}, aba^{-1}b^{-1}, a^{100}ba^{-100}b, \dots$$

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A word is *reducible* if it contains aa^{-1} or $a^{-1}a$ as a subword for any $a \in A$. Otherwise, it is *reduced*. We can now define

$$F(A) = \{w \in A^* \text{ reduced}\}$$

The group operation is concatenation, followed by reduction. For example,

$$(abab^{-1})(b^2a) = abab^{-1}b^2a = ababa$$

Note that reduction terminates as each reduction step reduces the length. We won't check that this is well defined or associative. The identity element is 1, inverses is clear.

A *presentation* consists of an alphabet A, which we will call *generators*, and a set $R \subseteq F(A)$, which we will call *relations*, and we write

$$\langle A \mid R \rangle = \langle a_1, a_2, \cdots \mid r_1, r_2, \ldots \rangle$$

This *presents* a group

$$G = \frac{F(A)}{\langle\langle R \rangle\rangle}$$

where $\langle\langle R \rangle\rangle$ is the *normal closure* of R, i.e. the smallest normal subrgoup of $F(A)$ containing R.

Example 1.1

Some examples of presentations which we have seen:

$$\begin{aligned}
\langle a \mid a^n \rangle &\cong \mathbb{Z}/n\mathbb{Z} \\
\langle r, s \mid r^n, s^2, srsr \rangle &\cong D_{2n} \\
\langle A \mid \rangle &\cong F(A) \\
\langle a, b \mid aba^{-1}b^{-1} \rangle &\cong \mathbb{Z}^2 \\
\langle a_1, \ldots, a_g, b_1, \ldots, b_g \mid a_1b_1a_1^{-1}b_1^{-1} \cdots a_gb_ga_g^{-1}b_g^{-1} \rangle &\cong \pi_1(\Sigma_g)
\end{aligned}$$

where Σ_g is the compact orientable surface of genus g.

As we see, presentations arise when we write down fundamental groups of spaces. In fact, all groups arise this way.

Corollary 1.2 (of Seifert-van Kampen)**.** For

$$G = \langle a_1, a_2, \cdots \mid r_1, r_2, \ldots \rangle$$

there exists a space X with $\pi_1(X) = G$.

Proof. First, start of with a wedge of circles, one for each a_i. Also consider a disjoint union of discs, one for each r_i. Attach the i-cell along its boundary, which is a loop in the wedge of circles. □

This is called the *presentation complex* of G. For example, if we have $\mathbb{Z}^2 = \langle a, b \mid aba^{-1}b^{-1} \rangle$, then $X = \mathrm{T}^2$ is the 2-torus.

In 1911, Max Dehn posed the following problems.

1. (Word problem) Given $w \in A^*$, determine whether or not $w = 1$ in $G = \langle A \mid R \rangle$. That is, whether or not $w \in \langle\langle R \rangle\rangle \subseteq F(A)$.
2. (Conjugacy problem) Given $G = \langle A \mid R \rangle$, $u, v \in A^*$, determine whether or not u is conjugate to v in G.
3. (Isomorphism problem) Given $G = \langle A \mid R \rangle$, $H = \langle B \mid S \rangle$, determine whether or not G is isomorphic to H.

Lecture 2

Remark 1.3. The conjugacy problem is stronger than the word problem, since $w = 1$ if and only if w is conjugate to 1.

Dehn was motivated by topology, but the problems asks for algorithms. We will often solve them using geometry.

All three were *un*solved in the 1950s, as all three problems are algorithmically undecidable. Nevertheless, positive solutions are known for many reasonable classes of groups.

Let $A = \{a_1, \dots, a_n\}$ be a finite alphabet.

Example 1.4 (word problem in free groups)

Let $w \in A^*$. If w is reduced, then $w = 1$ if and only if $w = \varnothing$. Otherwise, w contains a subword aa^{-1} for some $a \in A \cup A^{-1}$. Cancelling aa^{-1} produces a word $w' \in A^*$, such that $w = w'$ in $F(A)$, and $\ell(w') = \ell(w) - 2 < \ell(w)$. This terminates after finitely many steps.

We can also solve the conjugacy problem for free groups.

Definition 1.5

There is a natural action of $\mathbb{Z}$ on A^* permuting words. That is,

$$1 \cdot a_1 \cdots a_k = a_2 a_3 \cdots a_k a_1$$

The elements of $\mathbb{Z} \cdot w$ are called the *cyclic conjugates* of w. Note that all $u \in \mathbb{Z}w$ are conjugate to w. The orbits $\mathbb{Z}\backslash A^*$ are called *cyclic words*. A word is *cyclically reduced* if every cyclic conjugate is reduced.

Example 1.6

aba^{-1} is reduced, but not cyclically reduced as $ba^{-1}a$ is not reduced.

Note that if w is reduced, but not cyclically reduced, then

$$w = aw'a^{-1}$$

for some $a \in A \cup A^{-1}$. Note that w' is both conjugate to w and shorter than w. Hence after finitely many iterations, we can assume that w is cyclically reduced.

Lemma 1.7 (conjugacy problem in free groups)**.** If $u, v \in F(A)$ are cyclically reduced, then u is conjugate to v if and only if the corresponding cyclic words are the same.

Proof. If u, v have the same cyclic words, then v is a cyclic conjugate of u, which we have seen is a conjugate of u.

Conversely, suppose $u = gvg^{-1}$. By induction on $\ell(g)$, we can assume $g = a \in A \cup A^{-1}$. From this it follows that either $v = a^{-1}v'$, or $v = v'a$, as v is cyclically reduced. That is, $u = v'a^{-1}$ or $u = av'$. In both cases, they are cyclic conjugates. □

1.2 Historical case study

Let's briefly think about the state of topology in the early 20th century. Poincaré knew that homology classifies the compact two-dimensional surfaces. This motivated the

Conjecture 1.8 (Poincaré conjecture, version 1)**.** Let M be a compact connected 3-manifold, with

$$\mathrm{H}_*(M) = \begin{cases} \mathbb{Z} & * = 0, 3 \\ 0 & \text{otherwise} \end{cases}$$

Then M is homeomorphic to S^3.

Such a 3-manifold M is called a *homology sphere*.

Theorem 1.9 (Poincaré, 1904)**.** There exists a three dimensional homology sphere P, with

$$\pi_1(P) \twoheadrightarrow A_5$$

The moral is that: homology is not enough, we need use π_1 as well.

Conjecture 1.10 (Poincaré conjecture, version 2)**.** Let M be a compact connected 3-manifold, with $\pi_1(M) = 1$. Then M is homeomorphic to S^3.

This was proven by Perelman in 2003. Returning to the original conjecture, in 1910 Dehn wanted to construct more homology spheres.

Theorem 1.11 (Dehn, 1910)**.** There are infinitely many non-homeomorphic 3-dimensional homology spheres.

Remark 1.12. The isomorphism problem is exactly what is needed to distinguish these manifolds.

Lecture 3

Here is Dehn's construction. Consider the trefoil knot $K \subseteq \mathbb{R}^3 \subseteq S^3$. Let $N(K)$ be a regular open neighbourhood of K, i.e. a thickening of K. Let $N = S^3 \setminus N(K)$. This is a compact 3-manifold with boundart $\partial N \cong \mathrm{T}^2$.

Computing,

$$\mathrm{H}_*(N) = \begin{cases} \mathbb{Z} & * = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\pi_1(N) \cong \langle x, y, z \mid x^2 = y^3 = z \rangle$$

This follows from the fact that K is a torus knot, and we can compute the π_1 of the complement of a torus knot. Note that the abelianisation map

$$\pi_1(N) \to \mathrm{H}_1(N) \cong \mathbb{Z}$$

sends z to 6, x to 3 and y to 2. Moreover, the boundary torus T has $\pi_1(T) = \mathbb{Z}^2$, generated by xy and z. Under the abelianisation map, xy is sent to 5.

We can glue a solid torus $D^2 \times S^1$ to N, by a homeomorphism on the boundary. Let λ, μ be the corresponding loops on T^2. The resulting manifold $M_\phi = N \cup_\phi T$ is closed. $\pi_1(M_\phi)$ depends on $g = \phi_*(\mu)$. By Seifert-van Kapmen,

$$\pi_1(M_\phi) = \frac{\pi_1(N)}{\langle\langle g \rangle\rangle}$$

Similarly,

$$\mathrm{H}_1(M_\phi) = \frac{\mathbb{Z}}{\langle [g] \in \mathrm{H}^1(N) \rangle}$$

To produce a homology sphere, we need to choose ϕ such that $g = \phi_*(\mu) = 1$ in $\mathrm{H}^1(N)$.

If $g = (xy)^a z^b$, then in $\mathrm{H}^1(N)$ this is mapped to $5a + 6b \in \mathbb{Z}$. Choose $a = 6n + 5$, and $b = -(5n + 4)$ for $n \in \mathbb{Z}$. He constructs ϕ_n such that

$$\phi_n(\mu) = g_n = (xy)^{6n+5} z^{-(5n+4)}$$

His family of manifolds

$$D_n = N \cup_{\phi_n} U$$

has

$$\pi_1(D_n) = \langle x, y, z \mid x^2 = y^2 = z, (xy)^{6n+5} = z^{5n+4} \rangle$$

The remaining challenge is to prove the groups $G_n = \pi_1(D_n)$ for $n \geq 0$ are pairwise non-isomorphic.

This is the isomorphism problem! In particular,

$$g_n = g_m \implies g_n \text{ and } g_m \text{ are conjugate} \implies G_n \cong G_m$$

So we also need to solve the word and conjugacy problem in $\pi_1(N)$.

1.3 van Kampen diagrams

Definition 1.13

A map $f : Y \to X$ of cell complexes is called *combinatorial* if for all $k \in \mathbb{Z}_{\geq 0}$, and every k-cell e^k of Y, f maps the interior $\mathrm{Int}(e^k)$ homeomorphically to the interior of a k-cell of X.

Consider a presentation $G = \langle a_i \mid r_j \rangle$, and the associated presentation complex X.

Definition 1.14

A *(singular) disc diagram* is a compact contractible 2-complex D, with an embedding $D \hookrightarrow \mathbb{R}^2$. A disc diagram D is *over* X if it is equipped with a combinatorial map $D \to X$.

Example 1.15

Example of disc diagram.

Recall that X is given by a wedge of circles, with discs glued on for each relation. So the 1-cells correspond to generators, 2-cells go to relations (or cyclic conjugates, or inverses). With this:

- every oriented 1-cell of D is labelled with some $a_i \in A$,
- so that each 2-cell has boundary which is a cyclic conjugate of some $r_j^{\pm 1}$.

Associated to each disc diagram D, we have a *boundary cycle*, which reads a (cyclic[1]) word $w \in A^*$, which reduces to an element $w' \in \langle\langle r_j \rangle\rangle \leq F(A)$. To see this, D is contractible.

D is a *van Kampen diagram for* w.

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Lemma 1.16 (van Kampen)**.** If $w \in \langle\langle R \rangle\rangle$, then there exists a van Kampen diagram for w.

Proof. Suppose $w \in \langle\langle R \rangle\rangle$. Then w can be written as

$$w = \prod_{i=1}^{k} h_i r_i^{\pm 1} h_i$$

in $F(A)$, where $h_i \in F(A)$, and $r_i \in R$. Now build a *lollipop diagram* D_0, which has boundary word w_0, which is equal to w in $F(A)$, but may not be reduced.

If w_0 is reduced, $w = w_0$, and so we are done. Otherwise, w_0 contains a cancelling pair, so

$$w_0 = \cdots a a^{-1} \cdots$$

for some $a \in A \cup A^{-1}$. In w_0, this looks like:

We can see that e_1, e_2 share a vertex. There are two cases to consider:

1. if the *origin* of e_1 is the *terminus* of e_2, then the diagram D_0 is a wedge $D_1 \vee D'$. Then D_1 is a van Kampen diagram for w_1, which is the result of cancelling a and a^{-1}.
2. if the origin of e_1 is distinct from the terminus of e_2, then we can fold the edges to get D_1, which has boundary word w_1 as above.

In either case, $w_1 = \partial D_1$ is obtained from w_0 by cancelling a pair. Therefore, we may proceeed by induction, and after finitely many repetitions, we construct a van Kampen diagram D_n such that $w_n = \partial D_n$ is reduced, and $w_n = w$ in $F(A)$. Thus, $w_n = w$ as words, and so D_n is a van Kampen diagram for w. □

[1] or a word once we choose a base point.

Remark 1.17. The minimal number of 2-cells in a van Kampen diagram of w is the minimal number of k, such that w can be written as

$$w = \prod_{i=1}^{k} h_i r_i^{\pm 1} h_i^{-1}$$

This is called the *area* of w.

Example 1.18

Let $G = \mathbb{Z}^2 = \langle a, b \mid [a, b] \rangle$. Consider $w = a^n b^n a^{-n} b^{-n}$. This has van Kampen diagram

In this case, $\text{Area}(D) = n^2$. We will show D is minimal, and so $\text{Area}(w) = n^2$.

Definition 1.19

Let $\mathcal{P} = \langle A \mid R \rangle$ be a finite presentation of a group G. Define

$$\begin{aligned} \delta_{\mathcal{P}} : \mathbb{N} &\to \mathbb{N} \\ \ell &\mapsto \max_{w \in \langle\langle R \rangle\rangle, \ell(w) = d} \text{Area}(w) \end{aligned}$$

This is called the *Dehn function*.

Remark 1.20. The word problem in $\mathcal{P}$ if and only if $\delta_{\mathcal{P}}$ is computable.

2 Basics of geometric group theory

2.1 Cayley graphs

A *graph* is a 1-dimensional cell complex. Throughout, let G be a group, with finite generating set $S \subseteq G$.

Definition 2.1

The *Cayley graph* $\text{Cay}_S(G)$ is defined as follows:

- vertices $V(\text{Cay}_S(G)) = G$,
- edges $E(\text{Cay}_S(G))$ correspond bijectively with $G \times S$. That is, we have an edge $g \to gs$.

Example 2.2

The trivial group given by $1 = \langle a, b \mid a, b \rangle$ has Cayley graph

Example 2.3

$S_3 = \langle r, s \mid srsr, r^3, s^2 \rangle$ has Cayley graph

Example 2.4

$\mathbb{Z} = \langle 1 \rangle$ as Cayley graph

On the other hand, $\mathbb{Z} = \langle 2, 3 \rangle$ has Cayley graph

Note that the action of G on itself on the *left* extends to an action of G on $\text{Cay}_S(G)$, sending an edge $h \to hs$ to $gh \to ghs$. Note the right action does not work, because of our definition of the Cayley graph.

Remark 2.5. The action of G on $\mathrm{Cay}_S(G)$ is free. That is, for all $x \in \mathrm{Cay}_S(G)$, $\mathrm{Stab}_G(x) = 1$.

Proposition 2.6. Let $G = \langle S \mid R \rangle$, and let X be the corresponding presentation complex. Then there exists a G-equivariant isomorphism of graphs

$$\mathrm{Cay}_S(G) \cong \widetilde{X}_{(1)}$$

with the 1-skeleton of the universal cover $\widetilde{X}$ of X[a].

[a] and not the other way around.

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Proof. Consider the natural free action of $G = \pi_1(X)$ on $\widetilde{X}$, by deck transformations. The action is by combinatorial endomorphisms. In particular, it preserves the 1-skeleton. So we have a free action of G on $\widetilde{X}_{(1)}$, which sends vertices to vertices and edges to edges.

The action of G on $\widetilde{X}_{(0)}$ is free, and as X has only one vertex, transitive. Therefore, choosing a base vertex $\widetilde{v}_0$, the orbit-stabiliser theorem provides a G-equivariant bijection

$$G \to \mathrm{Orb}(\widetilde{v}_0) = \widetilde{X}_{(0)}$$

sending g to $g \cdot \widetilde{v}_0$. So this matches up the vertices as required.

Next, let us match up the edges. For each $s \in S$, let e_s be the corresponding edge of X. Let $\widetilde{e}_s$ be the unique lift of e_s to $\widetilde{X}$, beginning at $\widetilde{v}_0$. By the definition of the action of G on $\widetilde{X}$, e_s ends at $s \cdot \widetilde{v}_0$. Now an arbitrary edge $\widetilde{e}$ of $\widetilde{X}$ maps to some e_s, under the covering map. Since egdes of X correspond to G-orbits of edges in $\widetilde{X}$, it folloes that $\widetilde{e} = g \cdot \widetilde{e}_s$ for some $s \in S$. That is, $\widetilde{e}$ is the edge from $g \cdot \widetilde{v}_0$ to $gs \cdot \widetilde{v}_0$. So it corresponds to an edge from g to gs.

This shows that the G-equivariant map $G \to \widetilde{X}_{(0)}$ extends to a G-equivariant isomorphism of graphs as claimed. □

The next proposition deepens the relationship between generating sets and path connectedness.

Proposition 2.7. Let $\widetilde{X}$ be a path connected topological space, and suppose that G acts on $\widetilde{X}$ by homeomorphisms. If $U \subseteq \widetilde{X}$ is an open subset, such that $G \cdot U = \widetilde{X}$, then the set

$$S = \{g \in G \mid g \cdot U \cap U \neq \varnothing\}$$

generates G.

Proof. Fix a base point $\widetilde{x}_0 \in U$. Now for $g \in G$, let $\gamma : [0,1] \to \widetilde{X}$ be a path from $\widetilde{x}_0$ to $g \cdot \widetilde{x}_0$. The set $\{\gamma^{-1}(h \cdot U) \mid h \in G\}$ is an open cover of $[0,1]$. So it has a finite subcover, $\{\gamma^{-1}(U_1), \ldots, \gamma^{-1}(U_n)\}$, where $U_i = g_i \cdot U$. We may choose the indices so that

- $\widetilde{x}_0 \in U_1$,
- $\gamma^{-1}(U_i) \cap \gamma^{-1}(U_{i+1}) \neq \varnothing$ for all i,
- $g \cdot \widetilde{x}_0 \in U_n$.

Note that the g_i need not be unique. By definition, $x_0 \in U \cap g_1 \cdot U$, and so $g_1 \in S$. Similarly, if $t_i \in \gamma^{-1}(U_i) \cap \gamma^{-1}(U_{i+1})$, then $x_i = \gamma(t_i) \in g_i \cdot U \cap g_{i+1} \cdot U$. Thus,

$$g_i^{-1} \cdot x_i \in U \cap g_i^{-1} g_{i+1} \cdot U$$

and so $s_i = g_i^{-1} g_{i+1} \in S$. Thus, $g_n = s_{n-1} \cdots s_2 s_1 g_1$, is a finite product of elements of S. Finally, $g^{-1} g_n \in S$ similarly to the above, so $g \in \langle S \rangle$ as required. □

Example 2.8

Let $\Gamma \subseteq \mathrm{Isom}(\mathbb{R}^2)$ be the symmetry group of the standard tiling of the plane by equilateral triangles. Let U be a thickened triangle. Using the proposition, we obtain a finite generating set of Γ. In particular, Γ is generated by the reflections in the sides of a single triangle.

In particular, this is not a covering space action, as it is not free.

Definition 2.9

An action of G on $\widetilde{X}$ by homeomorphisms is *properly discontinuous* if for every compact $K \subseteq \widetilde{X}$, the set

$$\{g \cdot K \cap K\}$$

is finite.

The action is *cocompact* if there exists $K \subseteq \widetilde{X}$ compact, such that

$$G \cdot K = \widetilde{X}$$

$\widetilde{X}$ is *locally compact* if for every neighbourhood U of x, there exists an open neighbourhood $V \subseteq U$ of x, such that $\widetilde{V} \subseteq U$ is compact.

Corollary 2.10. If G acts on $\widetilde{X}$ properly discontinuously and cocompactly, and $\widetilde{X}$ is path connected and locally compact, then G is finitely generated.